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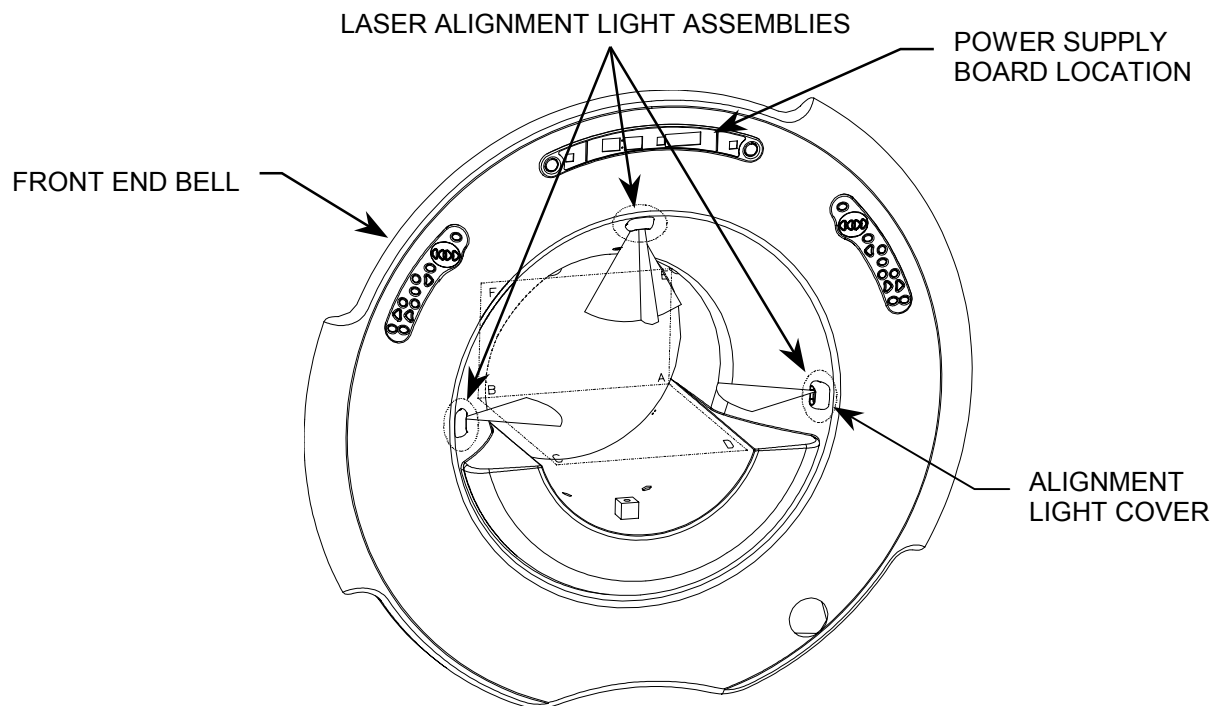
REVISION HISTORY 7

Description - Procedures for replacing alignment lights, and alignment of the laser lights. There are two (2) laser lights in the top middle assembly and one (1) each in the side assemblies.

1- LASER LIGHT REPLACEMENT

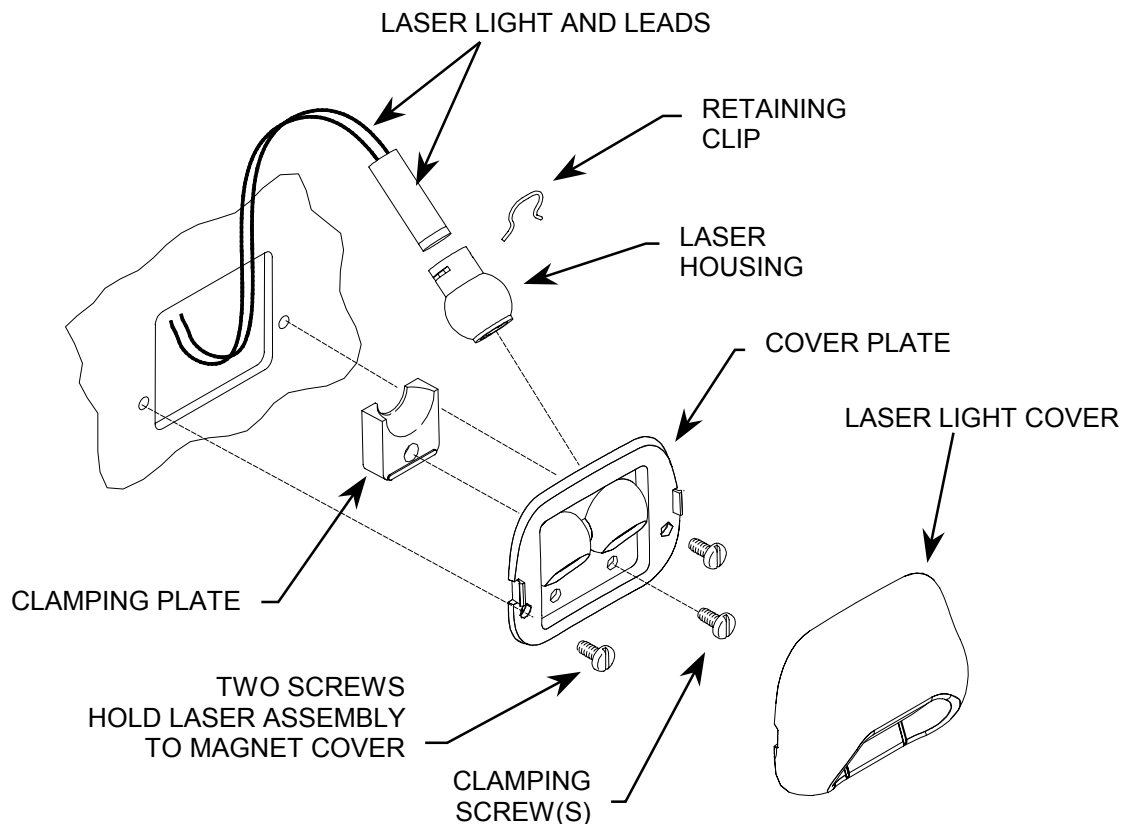
1-1 Laser Light Removal

1. Perform Lock Out / Tag Out Process (refer to *MR Signa LX Service Methods CDROM*, Safety (SYSMSA2.DOC), Section 8- MR System Lock Out Procedures.).
2. Remove middle control panel to allow access to power supply board. Refer to procedure ME2REA10.DOC Control Panel Replacement.
3. Pry cover off of defective laser assembly with a screwdriver. See Illustration 1-1.



ALIGNMENT LIGHT LOCATIONS
ILLUSTRATION 1-1

4. Remove two screws, and let laser light assembly hang free. See Illustration 1-2.
5. Remove screw clamping laser in place. See Illustration 1-2.
6. Using a needle nose pliers, pull retaining clip off laser/housing assembly. See Illustration 1-2.
7. Slide laser light out of housing. See illustration 1-2.

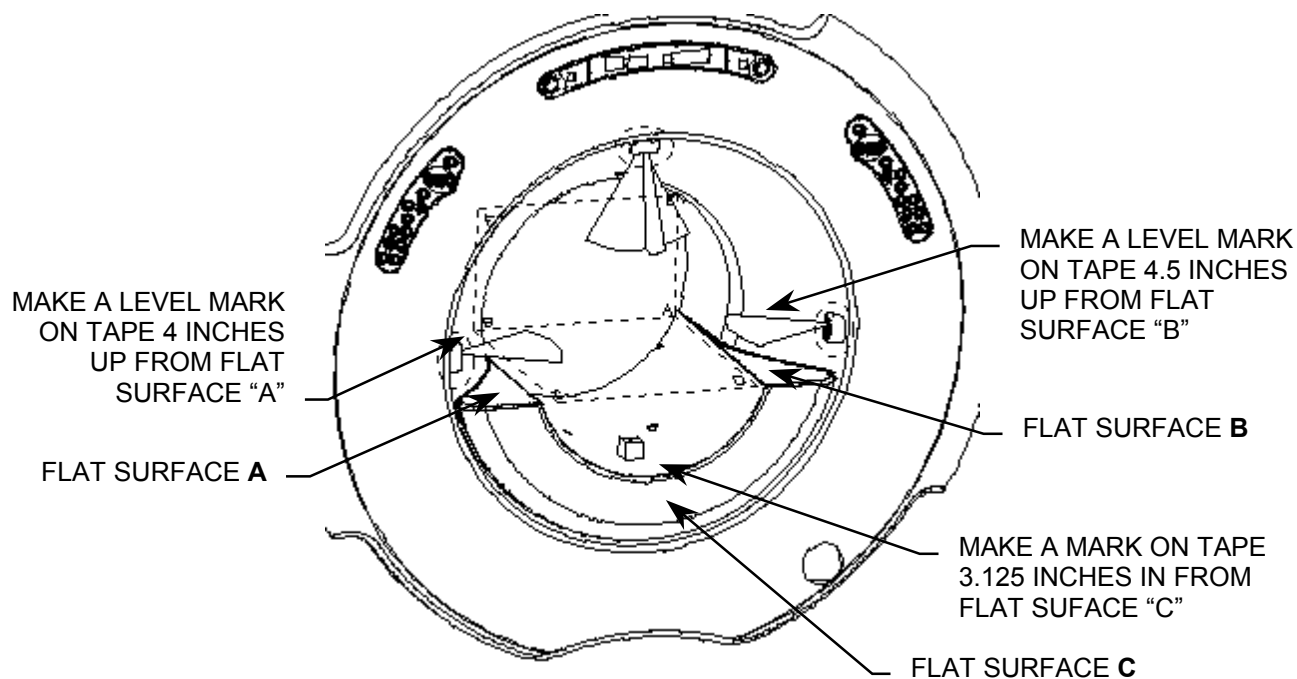


LASER ALIGNMENT LIGHT ASSEMBLY
ILLUSTRATION 1-2

8. Follow laser light leads to the power supply board and pull connector off.

1-2 Laser Light Replacement

1. Plug new laser light into power supply board.
2. Slide housing onto laser and install retaining clip. See Illustration 1-2.
3. Install clamp and slotted screw onto cover plate. See Illustration 1-2.
4. Install cover plate to end bell using two furnished screws. See Illustration 1-2.
5. Replace center control panel.
6. Refer to the corresponding Lockout/Tagout power restoration process corresponding to the PDU in use as described in Section 8 – MR System Lockout Procedures (refer to SYMSA2.DOC) .
7. Turn on laser alignment lights and check alignment of replaced laser. If laser is not aligned, use Illustration 1-3 for alignment specifications. Rotate laser (see Illustration 2-1) to correct lasers projection, until it lines up with appropriate marks. Refer to Section 2 for procedure on how to properly align the laser lights.

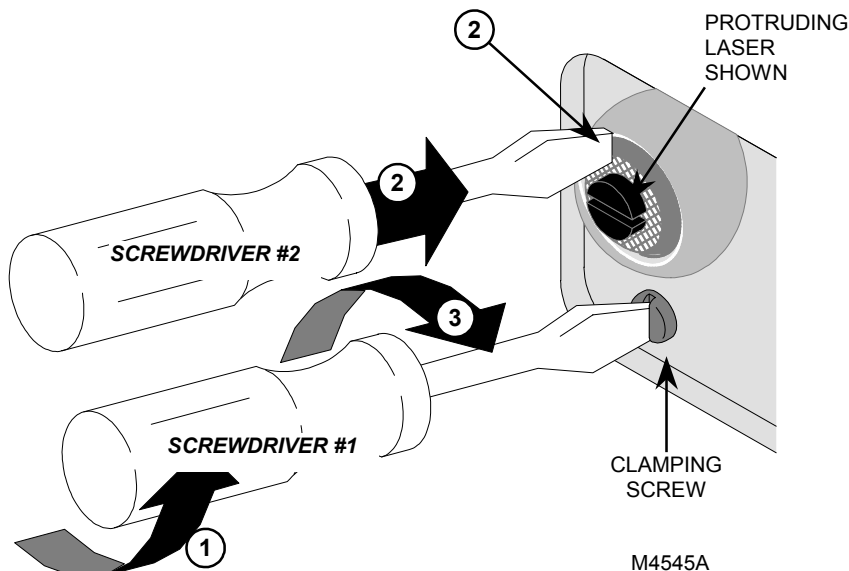
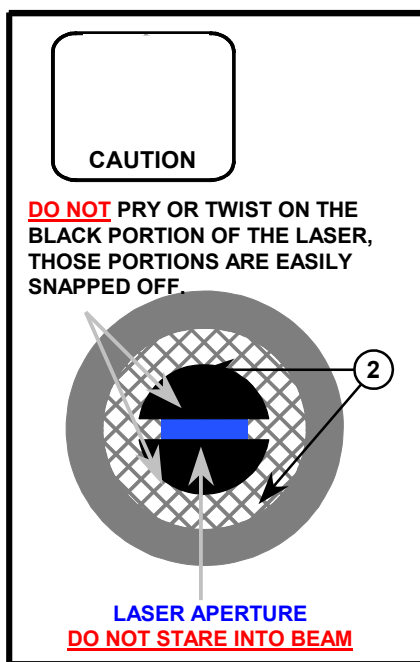


LASER ALIGNMENT
ILLUSTRATION 1-3

2 LASER LIGHT ALIGNMENT

2-1 Laser Light Alignment for Wide Open BRM/CRM Enclosures

NOTE:
LASERS WITH THEIR BLACK PORTION PROTRUDING PAST THE FACE
OF THE BRASS BALL (AS SHOWN), ARE PRONE TO BREAKAGE.
DO NOT USE THE BLACK PORTION TO PRY OR TWIST THE LASER.

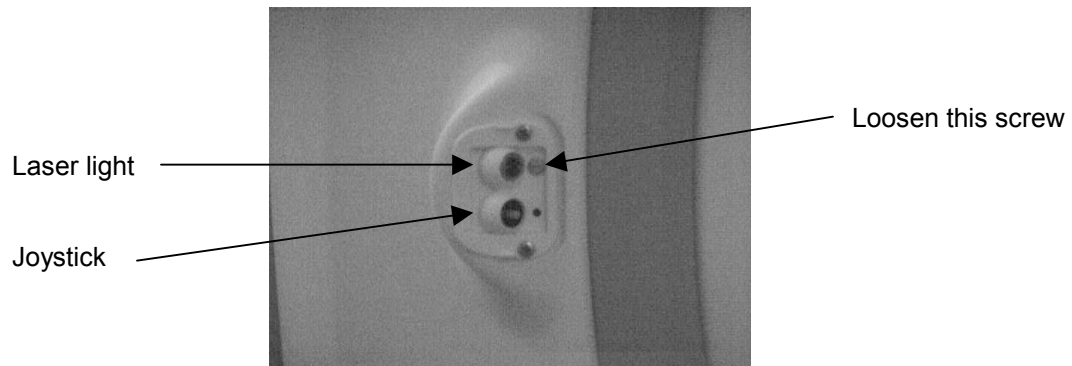


- ① **SCREWDRIVER #1;**
LOOSEN CLAMPING SCREW JUST ENOUGH TO ALLOW AXIAL MOVEMENT OF THE LASER BALL. KEEP CONTACT WITH THIS SCREW WHILE DOING THE AXIAL ADJUSTMENT (STEP 2). THEN TIGHTEN THE CLAMPING SCREW (STEP 3) BEFORE EITHER SCREWDRIVER IS REMOVED.
- ② **SCREWDRIVER #2;**
START AXIAL MOVEMENT OF THE LASER BALL WITH ONLY SLIGHT FORCE AND ONLY IN THE **CROSSHATCHED AREAS** (AS SHOWN).
- ③ **SCREWDRIVER #1;**
RETIGHTEN CLAMPING SCREW WHILE HOLDING LASER BALL IN POSITION WITH SCREWDRIVER #2.

LASER ALIGNMENT LIGHT ADJUSTMENT ILLUSTRATION 2-1

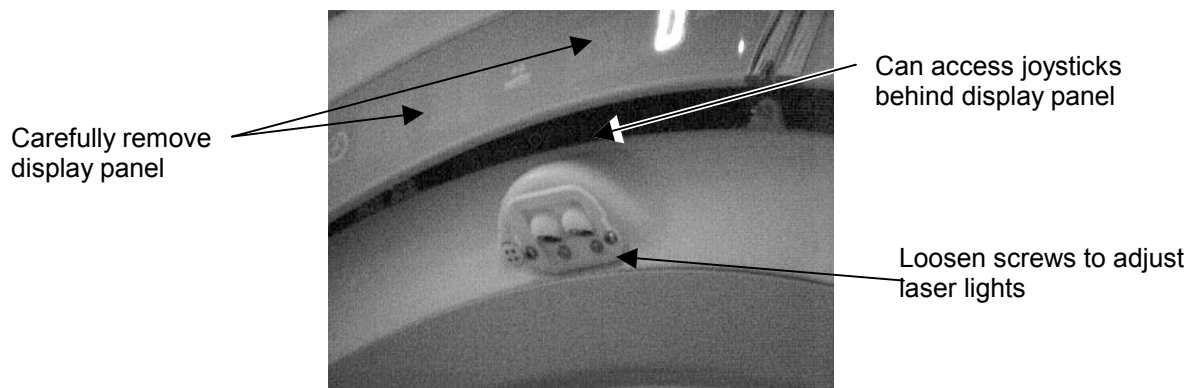
2-2 Laser Light Alignment for TwinSpeed Enclosures

1. Remove the Lens cover for the laser lights.
2. Turn the laser lights on.
3. Verify that the Coronal lights align with the light on the opposite side. If adjustment is needed, loosen the flat head screw, and move the "joystick" until the laser light is properly aligned. See Illustration 2-2.



CORONAL LIGHT ADJUSTMENT
ILLUSTRATION 2-2

4. Verify that the Axial and Sagittal laser lights are properly aligned. If adjustments are needed, remove the upper display by carefully prying the cover off. See Illustration 2-3. The joysticks for both the Sagittal and Axial can be accessed.



AXIAL & SAGITAL LIGHT ADJUSTMENT
ILLUSTRATION 2-3

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Nov 1, 1998	W. Pasciak	Initial revision
B	Dec 17, 1998	W. Pasciak	Correction to sheet 5
0	Feb 4, 1999	W. Pasciak	Final release
1	May 21, 1999	SM Atladottir	Updated Procedure References for New GUI
2	June 7, 2001	N. Nowakowski	1. Added text '1-2 Alignment Light Replacement' to TABLE OF CONTENTS Page 1. 2. Added Illustration 1-4, 'Laser Alignment Light Adjustment' procedure. 3. Updates and corrections throughout, particularly Sections 1-1 & 1-2. Lockout/Tagout clarification.
3	Feb 28, 2001	P. Kargard	Added alignment procedure for TwinSpeed Enclosure